CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 238519
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified

10/29/2025

Sold To	DNOW LP #CO02 540 Energy Park Dr Platteville, CO 80651	Ship To	DNOW LP #CO02 540 Energy Park Dr Platteville, CO 80651
Customer ID	34281		
TFA Sales Order	328265\1-1		
Purchase Order	26753318		

TFA Part ID	52311	Heat Code	A2472U
Description	6 STD CAP Y-52	Grade	Y52
Quantity	2	Raw Material Heat	A2G353
Customer Part ID	726951	Process	Y52-1GA
		Mill	SSAB

Austenitized @: 1750°	Tempered @: 1200°	Stress Relieved @:	Medium: OIL
Held For: 60 mins/min	Held For: 60 mins/min	Held For: mins/min	

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.18	1.03	0.012	0.002	0.24	0.04	0.12	0.15	0.30	<0.01	<0.01	0.0002	<0.01	0.04				0.42

Product Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.18	1.07	0.007	0.004	0.25	0.03	0.13	0.17	0.24	<0.01	<0.01	<0.0010	<0.01	0.04				0.42

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	*In-Process* Test 1	*In-Process* Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	67400 (psi)	85800 (psi)	30.50	78.00	4D	170			LONGITUDINAL	ROUND (.160" Diameter)

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

1. THIS MATERIAL WAS AUSTENITIZED AT1750F,OIL QUENCHED, AND TEMPERED AT 1200F. 2. THIS MATERIAL IS OF SEAMLESS MANUFACTURE. 3. IMPACTS ARE NOT REQUIRED FOR NPS 14 AND UNDER UNLESS SPECIFIC CUSTOMER SPEC REQUIRES THIS TEST.

1) MATERIAL MEETS MSS-SP-75 - 2019.
2) MATERIAL MEETS NACE MR-01-75/ISO 15156, REGIONS 1, 2 AND 3 - 2015 EDITION, ALSO MEETS NACE MR-0103 2015 EDITION.
3) MANUFACTURED IN AN ISO 9001 CERTIFIED FACILITY-CERTIFICATE #30248.
4) MATERIAL UNDER 16 NPS IS NOT IMPACT TESTED UNLESS ORDERED AS Y65 OR HIGHER YIELDS OR SUPPLEMENTARY REQUIREMENT (SR24) IS INVOKED PER MSS.

*UB = Unbroken Test

Heat: A2472U Grade: Y52

Scot Marshall, Manager Quality Assurance

The recording of fictitious entries on this document and or its fraudulent misuse may be punishable as a felony under Federal Statute.

Version 2.13.33.4

Job Number 257190

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		Mill	SSAB

Material Formed/Forged At: Tube Forgings of America Inc., Portland, OR

Material Heat Treated At: Met-Tek Inc., Clackamas, OR

TFA FITTINGS ARE MANUFACTURED IN PORTLAND, OR, USA. FOR THIS HEAT CODE, RAW MATERIAL SOURCE IS UNITED STATES

*UB = Unbroken Test

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